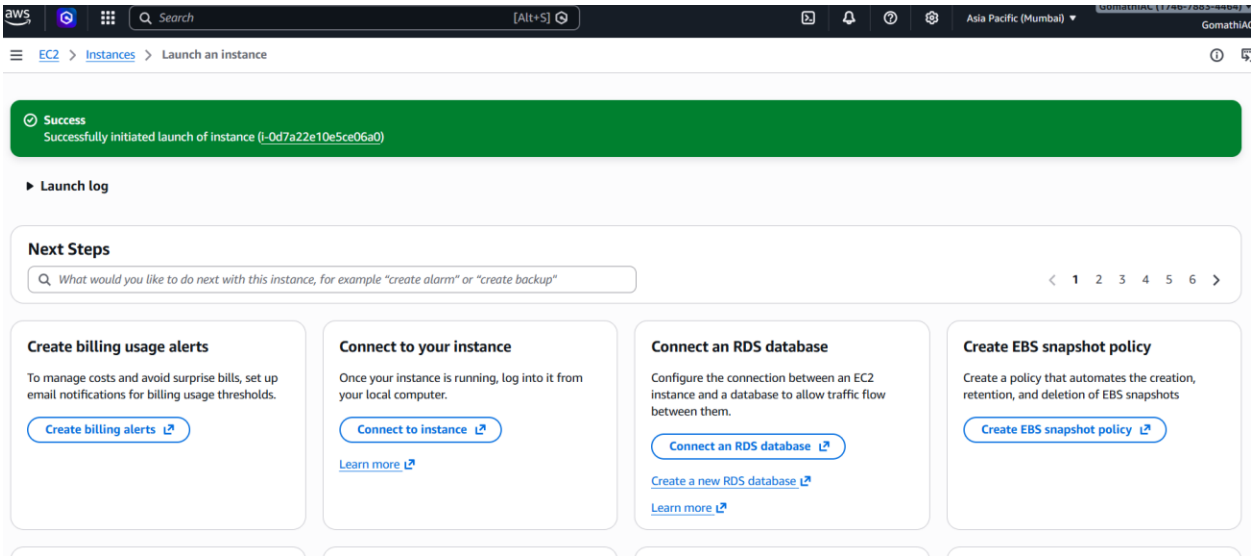


```
Using username "ec2-user".
Authenticating with public key "kuber"
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-172-31-45-172 ~]$ sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository   260 kB/s |  30 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-45-172 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:00:21 ago on Tue Mar 10 07:09:35 2026.
Dependencies resolved.
================================================================================
 Package                 Arch     Version                 Repository     Size
================================================================================
Installing:
 docker                  x86_64   25.0.14-1.amzn2023.0.2  amazonlinux    46 M
Installing dependencies:
 container-selinux       noarch   4:2.245.0-1.amzn2023    amazonlinux    58 k
 containerd              x86_64   2.2.1-1.amzn2023.0.1    amazonlinux    24 M
 iptables-libs           x86_64   1.8.8-3.amzn2023.0.2    amazonlinux   401 k
 iptables-nft            x86_64   1.8.8-3.amzn2023.0.2    amazonlinux   183 k
 libcgroup               x86_64   3.0-1.amzn2023.0.1      amazonlinux    75 k
 libnetfilter_conntrack  x86_64   1.0.8-2.amzn2023.0.2    amazonlinux    58 k
 libnfnetlink            x86_64   1.0.1-19.amzn2023.0.2   amazonlinux    30 k
 libnftnl                x86_64   1.2.2-2.amzn2023.0.2    amazonlinux    84 k
 pigz                    x86_64   2.5-1.amzn2023.0.3      amazonlinux    83 k
 runc                    x86_64   1.3.4-1.amzn2023.0.2    amazonlinux   3.9 M

Transaction Summary
================================================================================
Install  11 Packages

Total download size: 75 M
Installed size: 282 M
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.no 1.4 MB/s |  58 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_  13 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_6 5.2 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm 2.7 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023 2.0 MB/s |  58 kB     00:00
```

```
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/li
b/systemd/system/docker.socket.

  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch        11/11
  Running scriptlet: docker-25.0.14-1.amzn2023.0.2.x86_64                                                                                    11/
  Verifying        : container-selinux-4:2.245.0-1.amzn2023.noarch                                                                             1/
  Verifying        : containerd-2.2.1-1.amzn2023.0.1.x86_64                                                                                    2/
  Verifying        : docker-25.0.14-1.amzn2023.0.2.x86_64                                                                                      3/
  Verifying        : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                                                                                 4/
  Verifying        : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                                                  5/
  Verifying        : libcgroup-3.0-1.amzn2023.0.1.x86_64                                                                                       6/
  Verifying        : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                                                                        7/
  Verifying        : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                                                                                 8/
  Verifying        : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                                                                      9/
  Verifying        : pigz-2.5-1.amzn2023.0.3.x86_64                                                                                           10/
  Verifying        : runc-1.3.4-1.amzn2023.0.2.x86_64                                                                                         11/

Installed:
  container-selinux-4:2.245.0-1.amzn2023.noarch   containerd-2.2.1-1.amzn2023.0.1.x86_64   docker-25.0.14-1.amzn2023.0.2.x86_64                iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
  iptables-nft-1.8.8-3.amzn2023.0.2.x86_64        libcgroup-3.0-1.amzn2023.0.1.x86_64      libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64   libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64
  libnftnl-1.2.2-2.amzn2023.0.2.x86_64            pigz-2.5-1.amzn2023.0.3.x86_64           runc-1.3.4-1.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-172-31-45-172 ~]$ sudo service docker start
Redirecting to /bin/systemctl start docker.service
[ec2-user@ip-172-31-45-172 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-172-31-45-172 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-45-172 ~]$ docker run hello world
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.so
 connect: permission denied.
```

```
[ec2-user@ip-172-31-37-101 ~]$ sudo usermod -aG docker $USER
[ec2-user@ip-172-31-37-101 ~]$ newgrp docker
[ec2-user@ip-172-31-37-101 ~]$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-172-31-37-101 ~]$ minikube start --driver=docker --memory=700mb --c
pus=1 --force
* minikube v1.38.1 on Amazon 2023.10.20260302
! minikube skips various validations when --force is supplied; this may lead to
unexpected behavior
* Using the docker driver based on user configuration

X Requested cpu count 1 is less than the minimum allowed of 2


X docker only has 916MiB available, less than the required 1800MiB for Kubernete
s


X System only has 916MiB available, less than the required 1800MiB for Kubernete
s


X Requested memory allocation 700MiB is less than the usable minimum of 1800MB


X Requested memory allocation (700MB) is less than the recommended minimum 1900M
B. Deployments may fail.


X The requested memory allocation of 700MiB does not leave room for system overh
ead (total system memory: 916MiB). You may face stability issues.
* Suggestion: Start minikube with less memory allocated: 'minikube start --memor
y=916mb'

! Starting v1.39.0, minikube will default to "containerd" container runtime. See
 #21973 for more info.
* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 68.53 M
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 15.13 M
```

```
ec2-user@ip-172-31-5-92 ~]$ ^C
ec2-user@ip-172-31-5-92 ~]$ sudo usermod -aG docker $USER
ec2-user@ip-172-31-5-92 ~]$ newgrp docker
ec2-user@ip-172-31-5-92 ~]$ docker version
Client:
 Version:           25.0.14
 API version:       1.44
 Go version:        go1.24.12
 Git commit:        0bab007
 Built:             Wed Dec 10 00:00:00 2025
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          25.0.14
  API version:      1.44 (minimum version 1.24)
  Go version:       go1.24.12
  Git commit:       d334795
  Built:            Wed Dec 10 00:00:00 2025
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          2.2.1+unknown
  GitCommit:
 runc:
  Version:          1.3.4
  GitCommit:        d6d73eb8c60246978da649ffe75ce5c8bca8f856
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
ec2-user@ip-172-31-5-92 ~]$ minikube start --driver=docker
* minikube v1.38.1 on Amazon 2023.10.20260302
* Using the docker driver based on user configuration
  Starting v1.39.0, minikube will default to "containerd" container runtime. See
#21973 for more info.

X The requested memory allocation of 1913MiB does not leave room for system over
```

```
* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 36.67 M
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 14.83 M
* Creating docker container (CPUs=2, Memory=1913MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namesp
ace by default
[ec2-user@ip-172-31-5-92 ~]$ kubectl cluster-info
Kubernetes control plane is running at https://192.168.49.2:8443
CoreDNS is running at https://192.168.49.2:8443/api/v1/namespaces/kube-system/se
rvices/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
[ec2-user@ip-172-31-5-92 ~]$ ^C
[ec2-user@ip-172-31-5-92 ~]$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   2m31s   v1.35.1
[ec2-user@ip-172-31-5-92 ~]$ kubectl get namespaces
NAME              STATUS   AGE
default           Active   2m43s
kube-node-lease   Active   2m43s
kube-public       Active   2m43s
kube-system       Active   2m44s
[ec2-user@ip-172-31-5-92 ~]$ kubectl get ns
NAME              STATUS   AGE
default           Active   6m29s
kube-node-lease   Active   6m29s
kube-public       Active   6m29s
kube-system       Active   6m30s
[ec2-user@ip-172-31-5-92 ~]$
```

```
  GNU nano 8.3                    namespace.yml
apiVersion: v1
kind: Namespace
metadata:
  name: test

                              [ Read 4 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut     ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste   ^J Justify   ^/ Go To Line
```

```
namespace/dev created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get ns
NAME              STATUS   AGE
default           Active   12m
dev               Active   26s
kube-node-lease   Active   12m
kube-public       Active   12m
kube-system       Active   12m
[ec2-user@ip-172-31-5-92 ~]$ kubectl run nginx --image=nginx -n dev
pod/nginx created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get pods -n dev
NAME    READY   STATUS    RESTARTS   AGE
nginx   1/1     Running   0          14s
[ec2-user@ip-172-31-5-92 ~]$ kubectl get all -n dev
NAME        READY   STATUS    RESTARTS   AGE
pod/nginx   1/1     Running   0          44s
[ec2-user@ip-172-31-5-92 ~]$ kubectl run nginx-pod \
--image=nginx \
--namespace=dev
pod/nginx-pod created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get pods -n dev
NAME        READY   STATUS    RESTARTS   AGE
nginx       1/1     Running   0          107s
nginx-pod   1/1     Running   0          10s
[ec2-user@ip-172-31-5-92 ~]$ nano namespace.yml
[ec2-user@ip-172-31-5-92 ~]$ nano namespace.yml
[ec2-user@ip-172-31-5-92 ~]$ kubectl apply -f namespace.yaml
error: the path "namespace.yaml" does not exist
[ec2-user@ip-172-31-5-92 ~]$ kubectl apply -f namespace.yml
namespace/test created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get ns
NAME              STATUS   AGE
default           Active   17m
dev               Active   4m57s
kube-node-lease   Active   17m
kube-public       Active   17m
kube-system       Active   17m
test              Active   11s
[ec2-user@ip-172-31-5-92 ~]$ kubectl config set-context --current --namespace=de
v
Context "minikube" modified.
[ec2-user@ip-172-31-5-92 ~]$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
nginx       1/1     Running   0          4m47s
nginx-pod   1/1     Running   0          3m10s
[ec2-user@ip-172-31-5-92 ~]$
```

```
kube-system       Active   12m
[ec2-user@ip-172-31-5-92 ~]$ kubectl run nginx --image=nginx -n dev
pod/nginx created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get pods -n dev
NAME    READY   STATUS    RESTARTS   AGE
nginx   1/1     Running   0          14s
[ec2-user@ip-172-31-5-92 ~]$ kubectl get all -n dev
NAME        READY   STATUS    RESTARTS   AGE
pod/nginx   1/1     Running   0          44s
[ec2-user@ip-172-31-5-92 ~]$ kubectl run nginx-pod \
--image=nginx \
--namespace=dev
pod/nginx-pod created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get pods -n dev
NAME        READY   STATUS    RESTARTS   AGE
nginx       1/1     Running   0          107s
nginx-pod   1/1     Running   0          10s
[ec2-user@ip-172-31-5-92 ~]$ nano namespace.yml
[ec2-user@ip-172-31-5-92 ~]$ nano namespace.yml
[ec2-user@ip-172-31-5-92 ~]$ kubectl apply -f namespace.yaml
error: the path "namespace.yaml" does not exist
[ec2-user@ip-172-31-5-92 ~]$ kubectl apply -f namespace.yml
namespace/test created
[ec2-user@ip-172-31-5-92 ~]$ kubectl get ns
NAME              STATUS   AGE
default           Active   17m
dev               Active   4m57s
kube-node-lease   Active   17m
kube-public       Active   17m
kube-system       Active   17m
test              Active   11s
[ec2-user@ip-172-31-5-92 ~]$ kubectl config set-context --current --namespace=de
v

Context "minikube" modified.
[ec2-user@ip-172-31-5-92 ~]$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
nginx       1/1     Running   0          4m47s
nginx-pod   1/1     Running   0          3m10s
[ec2-user@ip-172-31-5-92 ~]$ kubectl delete namespace test
namespace "test" deleted
[ec2-user@ip-172-31-5-92 ~]$ kubectl get all -n dev
NAME            READY   STATUS    RESTARTS   AGE
pod/nginx       1/1     Running   0          5m49s
pod/nginx-pod   1/1     Running   0          4m12s
[ec2-user@ip-172-31-5-92 ~]$ minikube stop
✋ Stopping node "minikube"  ...
🛑 Powering off "minikube" via SSH ...
🛑 1 node stopped.
[ec2-user@ip-172-31-5-92 ~]$
```